VUA-SEN307 DATABASE SYSTEMS — WEEK 4 PRACTICAL

Lab 2: Database Design and Normalization | Part 4, Question 1

1. Unnormalized table (UNF)

Registration(matric_no, student_name, student_dept, course_codes, course_titles, lecturers, lecturer_depts, session)

The original registration sheet contains multiple course codes, course titles, lecturers and lecturer departments in single cells, creating repeating groups.

2. First Normal Form (1NF)

Registration_1NF(matric_no, student_name, student_dept, course_code, course_title, lecturer, lecturer_dept, session)

Each row represents one student–course combination, and each cell contains one value. A possible composite key is (matric_no, course_code, session).

3. Second Normal Form (2NF)

Student(matric_no PK, student_name, student_dept)
Course(course_code PK, course_title)
Enrolment(matric_no PK/FK → Student, course_code PK/FK → Course, session)

Student details depend on matric_no, and course details depend on course_code, so these partial dependencies are separated from the student–course enrolment.

4. Third Normal Form (3NF)

Department(dept_id PK, dept_name)
Student(student_id PK, matric_no UNIQUE, full_name, dept_id FK → Department)
Lecturer(lecturer_id PK, lecturer_name, dept_id FK → Department)
Course(course_id PK, course_code UNIQUE, course_title)
Enrolment(student_id PK/FK → Student, course_id PK/FK → Course, session PK)
Teaching(lecturer_id PK/FK → Lecturer, course_id PK/FK → Course, session PK)

Department details are stored once in Department, and students and lecturers reference the department by foreign key. Teaching records which lecturer teaches which course in a given session, avoiding transitive dependencies and representing the many-to-many relationship.

Final schema key: PK = Primary Key; FK = Foreign Key; UNIQUE = value must not repeat. The Enrolment primary key is (student_id, course_id, session); the Teaching primary key is (lecturer_id, course_id, session).